Robin **Wroblowski**

Ph.D. Researcher in Neuromorphic Control & Spiking Neural Circuits

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Education

KU Leuven — Leuven, Belgium

Ph.D. in Neuromorphic Control — *2023 – Present*
- Advised by **Prof. Rodolphe Sepulchre** (supervisor) and **Prof. Benjamin Gorissen** (co-supervisor)

Master in Mathematical Engineering — *2021 – 2023*
- Graduated *cum laude*
- Exchange Semester at the Technical University of Vienna (Erasmus+, 2021)

Bachelor in Engineering Science — *2018 – 2021*
- Computer Science & Electrical Engineering
- Graduated *cum fructu*

Publications

Describing functions and phase response curves of excitable systems — European Control Conf. 2026
Robin Wroblowski, Rodolphe Sepulchre
- Published in the 2026 European Control Conference (ECC).

Neuromorphic design of a rhythmic network — Under Review 2026
Robin Wroblowski, Rodolphe Sepulchre
- Submitted to Journal of Neuromorphic Computing & Engineering.

On energy-based modelling for fluidic excitable neurons — Under Review 2026
Robin Wroblowski, Brent Michiels, Benjamin Gorissen, Rodolphe Sepulchre
- Submitted to the 2027 European Control Conference (ECC).

Regulating event-based interaction through rhythmic compliance — In Preparation
Robin Wroblowski, Rodolphe Sepulchre — Expected: Jan. 2027
- Target submission: January 2027, Automatica / Journal of Neuromorphic Computing and Engineering

CPG design for adaptive soft robotic locomotion and reflexive interaction — In Preparation
Robin Wroblowski, Brent Michiels, Benjamin Gorissen, Rodolphe Sepulchre — Expected: Apr. 2027
- Target submission: April 2027.

Other Experience

Vice-President — Leuven, Belgium
Vlaamse Technische Kring (VTK) — 2022 - 2023
- Directed internal relations and long-term strategy for the official engineering student association of 84 members representing over 4,000 students, and managed the daily operations across multiple organizational subdivisions.

IT Team Leader — Leuven, Belgium
Vlaamse Technische Kring (VTK) — 2020 - 2022
- Led the development and deployment of multiple software projects, including a large-scale ticketing service, a career profile matching platform and a logistics backend to support campus-wide events.

Council Member — Leuven, Belgium
International PhD Advisory Council (IPAC) — 2023 - Present
- Co-organized a yearly program of academic and social initiatives—including research seminars, game nights, and summer gatherings—to support the integration of international PhD students.

Skills

Scientific Computing	**Python**, **Matlab**, **Fortran**, Julia, Magma
Software/Hardware	**C++**, **Arduino**, **C**, PHP, Java, JavaScript
Spoken Languages	**Dutch (Native)**, **English (Native)**, **German (Advanced)**, French (Intermediate)

Presentations & Talks

Describing Functions and Phase Response Curves of Excitable Systems
European Control Conference (ECC) — July 2026

Regulating Rhythmic Compliance in Central Pattern Generators
45th Benelux Meeting on Systems and Control — March 2026

Rebound Amplifiers for Discrete-Event Pattern Generation
44th Benelux Meeting on Systems and Control — March 2025

Tuning the Entrainability of a Half-Center Oscillator
43rd Benelux Meeting on Systems and Control — March 2024

Doctoral Research & Teaching

Ph.D. Candidate in Neuromorphic Control — KU Leuven
Advised by Prof. Rodolphe Sepulchre — 2023 - Present

- **Design of Pattern Generating Networks:** A shift from coupled-oscillator networks to excitable networks yields a framework for distributed design and analysis of Central Pattern Generators based on event-propagation.
- **Regulating Rhythmic Interaction:** Developed methods for controlling the behavior of the event-based interconnection between a rhythmic plant and a neuromorphic controller by balancing the relative roles of feedforward and feedback.
- **Energy-based Modeling of Excitability:** Constructed an energy-based formulation of neuronal excitability to enable energetically consistent, domain-agnostic realizations of excitable networks (e.g., fluidic CPGs).

Teaching Assistant — KU Leuven
Faculty of Engineering Science — 2023 - Present

- Instructed and prepared materials for a Bachelor-level Systems & Control course over four years (serving as Lead Teaching Assistant for two years). Coordinated exercise sessions, designed exam questions, and managed grading.
- Provided daily supervision and mentorship to a Master's student for their thesis, "Neuromorphic Control of a Pendulum" demonstrating the practical verification of rhythmic neuromorphic control principles using a physical pendulum.

References

Prof. Rodolphe Sepulchre
PhD Advisor
Department of Electrical Engineering (STADIUS)
KU Leuven, Belgium
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Prof. Fulvio Forni
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Department of Engineering
University of Cambridge, United Kingdom
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Prof. Benjamin Gorissen
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Department of Mechanical Engineering (Soft Robotics)
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